

TAX ENGINE — FORMULAS & DATA REFERENCE

All calculation logic and parameter data for the ONYX Ledger engine · Tax years 2025 & 2026 · Prepared August 2026

How to use this: every formula here is deterministic — the engine is these formulas plus annual parameter tables. A companion file, `tax_engine_params.json`, holds the same numbers in machine-readable form so the code reads config, never hardcoded values. **Golden rule:** the 2025 figures apply to 2025 income (returns filed spring 2026); 2026 figures apply to income being earned now.

VERIFICATION STATUS (checked 13 August 2026): federal brackets (both years), the basic personal amount and its phase-out, Canada employment amount, age amount, medical threshold, disability amount, spouse amount, CPP, EI (federal and Quebec), RRSP/TFSA limits, the OAS clawback thresholds, capital-gains inclusion, dividend gross-up and federal dividend tax credit rates, and **all provincial and territorial brackets for 2025 and 2026** were confirmed line-by-line against CRA's published pages. Items still unverified are listed in §15 and flagged `"_verified": false` in the JSON.

This is not a compliance certification. Matching CRA's published numbers does not make software CRA-certified — certification is earned only by passing the CRA's own test returns during the NETFILE certification cycle.

1. THE CORE PIPELINE — HOW EVERY RETURN COMPUTES

```
totalIncome      = employment + selfEmployment + interest + grossedUpDividends
                  + taxableCapitalGains + otherIncome           // T1 line 15000
netIncome        = totalIncome - deductions                     // line 23600
                  // deductions: RRSP, FHSA, childcare, union dues,
                  // moving, support paid, carrying charges
taxableIncome    = netIncome - additionalDeductions             // line 26000
                  // e.g. losses carried forward, security options deduction

fedTaxBeforeCredits = applyBrackets(taxableIncome, FED_BRACKETS[year])
provTaxBeforeCredits = applyBrackets(taxableIncome, PROV_BRACKETS[prov][year])

fedTax = max(0, fedTaxBeforeCredits - fedNonRefundableCredits - dividendTaxCredit_fed)
provTax = max(0, provTaxBeforeCredits - provNonRefundableCredits - dividendTaxCredit_pro
              + provincialSurtax)                                // ON only, see §4

totalPayable = fedTax + provTax + CPP_selfEmployed_owed
refundOrOwing = amountsPaid - totalPayable
                // amountsPaid: tax withheld on T4/T5 slips (box 22),
                // instalments paid, refundable credits (CWB, etc.)
                // positive = refund, negative = balance owing
```

```
function applyBrackets(income, brackets):           // brackets sorted ascending
    tax = 0
    for each (threshold, rate, nextThreshold) in brackets:
        slice = min(income, nextThreshold) - threshold
        if slice <= 0: break
        tax += slice × rate
    return tax
```

2. FEDERAL TAX BRACKETS

Rate-cut note: the lowest federal rate was cut from 15% to 14% effective July 1, 2025 — so tax year **2025 uses a blended 14.5%** on the first bracket, and **2026 uses 14%**. Non-refundable credits are converted to tax value at this same lowest rate (see §3), so the credit rate is 14.5% for 2025 and 14% for 2026.

Bracket (taxable income)	2025 rate	2026 bracket	2026 rate
\$0 – \$57,375	14.5%	\$0 – \$58,523	14%
\$57,375 – \$114,750	20.5%	\$58,523 – \$117,045	20.5%
\$114,750 – \$177,882	26%	\$117,045 – \$181,440	26%
\$177,882 – \$253,414	29%	\$181,440 – \$258,482	29%
over \$253,414	33%	over \$258,482	33%

Annual indexation: thresholds and most credit amounts are multiplied by CRA's indexation factor each year (2025: 2.7% · 2026: 2.0%). Store the factor; every November CRA publishes next year's numbers.

3. FEDERAL NON-REFUNDABLE CREDITS (the big ones)

All credits below are **amounts**; convert to tax reduction via: `creditValue = amount × lowestFedRate` (14.5% in 2025, 14% in 2026). Credits can reduce tax to zero but never below (except refundable ones, §9).

3.1 Basic Personal Amount (BPA) — with high-income phase-out

```
// 2025: max 16,129, min 14,538, phase-out range = 4th bracket (177,882 → 253,414)
// 2026: max 16,452, min ~14,829 (verify), range 181,440 → 258,482
if netIncome <= rangeStart:      bpa = bpaMax
else if netIncome >= rangeEnd:    bpa = bpaMin
else: bpa = bpaMax - (bpaMax - bpaMin) × (netIncome - rangeStart) / (rangeEnd - rangeSta
```

3.2 The rest of the standard stack

✓ = confirmed against CRA's indexation page, 13 Aug 2026. ? = not yet verified.

Credit	2025	2026	Formula / rule
Canada Employment Amount ✓	\$1,471	\$1,501	$\min(\text{amount}, \text{employmentIncome})$ — everyone with a T4 gets it
CPP/QPP contributions ✓	actual paid		base CPP contributions are a credit; the enhanced portion (incl. CPP2) is a deduction from income instead — see §5
EI premiums ✓	max \$1,077.48	max \$1,123.07	credit for premiums from T4 box 18
Age amount (65+) ✓	\$9,028	\$9,208	reduced by 15% of netIncome over the threshold, to zero. Threshold itself unverified (2025 believed \$45,522)
Medical expenses ✓	cap \$2,834	cap \$2,890	$\max(0, \text{eligibleMedical} - \min(0.03 \times \text{netIncome}, \text{cap}))$ — claim on either spouse; the lower-income spouse usually wins . 12-month claim window ending in tax year
Disability (DTC) ✓	\$10,138	\$10,341	requires CRA-approved T2201; transferable from dependant
Spouse / common-law partner ✓	\$16,129 (\$18,816 w/ caregiver)	\$16,452 (\$19,192 w/ caregiver)	reduced dollar-for-dollar by the dependant's net income
Charitable donations ?	two-tier		$\text{lowestRate} \times \min(200, \text{don}) + 29\% \times \text{amountOver200}$ (33% on the portion matching income in the 33% bracket). Carry forward 5 years — optimizer opportunity: pooling . Tier rates post-rate-cut not confirmed
Tuition (T2202) ?	actual		credit at lowest rate; unused carries forward indefinitely or transfers up to \$5,000 to parent/spouse
Home buyers' amount ?	\$10,000		flat statutory amount (not indexed), year of first-home purchase — not shown on the indexation page

4. PROVINCIAL & TERRITORIAL TAX — ALL 12 JURISDICTIONS, BOTH YEARS

Same `applyBrackets` function, different tables. **All brackets and rates below were confirmed against CRA's current-year and last-year pages on 13 August 2026.** Quebec excluded (separate TP1 system via Revenu Québec).

4.1 — 2025 brackets (used on returns filed spring 2026)

Jurisdiction	2025 brackets → rates
Alberta	to \$60,000: 8% · to \$151,234: 10% · to \$181,481: 12% · to \$241,974: 13% · to \$362,961: 14% · above: 15%
British Columbia	to \$49,279: 5.06% · to \$98,560: 7.7% · to \$113,158: 10.5% · to \$137,407: 12.29% · to \$186,306: 14.7% · to \$259,829: 16.8% · above: 20.5%
Manitoba	to \$47,564: 10.8% · to \$101,200: 12.75% · above: 17.4%
New Brunswick	to \$51,306: 9.4% · to \$102,614: 14% · to \$190,060: 16% · above: 19.5%
Newfoundland & Labrador	to \$44,192: 8.7% · to \$88,382: 14.5% · to \$157,792: 15.8% · to \$220,910: 17.8% · to \$282,214: 19.8% · to \$564,429: 20.8% · to \$1,128,858: 21.3% · above: 21.8%
Northwest Territories	to \$51,964: 5.9% · to \$103,930: 8.6% · to \$168,967: 12.2% · above: 14.05%
Nova Scotia	to \$30,507: 8.79% · to \$61,015: 14.95% · to \$95,883: 16.67% · to \$154,650: 17.5% · above: 21%
Nunavut	to \$54,707: 4% · to \$109,413: 7% · to \$177,881: 9% · above: 11.5%
Ontario	to \$52,886: 5.05% · to \$105,775: 9.15% · to \$150,000: 11.16% · to \$220,000: 12.16% · above: 13.16%
Prince Edward Island	to \$33,328: 9.5% · to \$64,656: 13.47% · to \$105,000: 16.6% · to \$140,000: 17.62% · above: 19%
Saskatchewan	to \$53,463: 10.5% · to \$152,750: 12.5% · above: 14.5%
Yukon	to \$57,375: 6.4% · to \$114,750: 9% · to \$177,882: 10.9% · to \$500,000: 12.8% · above: 15%

4.2 — 2026 brackets (current income year)

Jurisdiction	2026 brackets → rates
Alberta	to \$61,200: 8% · to \$154,259: 10% · to \$185,111: 12% · to \$246,813: 13% · to \$370,220: 14% · above: 15%
British Columbia	to \$50,363: 5.6% · to \$100,728: 7.7% · to \$115,648: 10.5% · to \$140,430: 12.29% · to \$190,405: 14.7% · to \$265,545: 16.8% · above: 20.5%
Manitoba	unchanged from 2025 — to \$47,564: 10.8% · to \$101,200: 12.75% · above: 17.4%
New Brunswick	to \$52,333: 9.4% · to \$104,666: 14% · to \$193,861: 16% · above: 19.5%
Newfoundland & Labrador	to \$44,678: 8.7% · to \$89,354: 14.5% · to \$159,528: 15.8% · to \$223,340: 17.8% · to \$285,319: 19.8% · to \$570,638: 20.8% · to \$1,141,275: 21.3% · above: 21.8%
Northwest Territories	to \$53,003: 5.9% · to \$106,009: 8.6% · to \$172,346: 12.2% · above: 14.05%

Nova Scotia	to \$30,995: 8.79% · to \$61,991: 14.95% · to \$97,417: 16.67% · to \$157,124: 17.5% · above: 21%
Nunavut	to \$55,801: 4% · to \$111,602: 7% · to \$181,439: 9% · above: 11.5%
Ontario	to \$53,891: 5.05% · to \$107,785: 9.15% · to \$150,000: 11.16% · to \$220,000: 12.16% · above: 13.16%
Prince Edward Island	6 brackets now: to \$33,928: 9.5% · to \$65,820: 13.47% · to \$106,890: 16.6% · to \$142,520: 17.62% · to \$200,000: 19% · above: 20% (new)
Saskatchewan	to \$54,532: 10.5% · to \$155,805: 12.5% · above: 14.5%
Yukon	mirrors federal thresholds — to \$58,523: 6.4% · to \$117,045: 9% · to \$181,440: 10.9% · to \$500,000: 12.8% · above: 15%

Three things to note in the 2026 tables: (1) **British Columbia's lowest rate reads 5.6% for 2026 versus 5.06% for 2025** — that is a rate change, not indexation. Re-confirm on BC's own page before this touches a user's number. (2) **PEI added a sixth bracket** with a new 20% top rate above \$200,000. (3) **Manitoba did not index** — its 2026 thresholds are identical to 2025.

Still missing for provincial tax: provincial basic personal amounts, provincial dividend tax credit rates, and Ontario's surtax thresholds. Provincial tax cannot be computed correctly without these — source them per province before shipping.

4.1 Ontario surtax — the one weird extra formula

```
// applied AFTER Ontario credits, on Ontario basic tax:
surtax = 0.20 × max(0, onBasicTax - 5710) + 0.36 × max(0, onBasicTax - 7307) // 2025 t
onTax = onBasicTax + surtax - ontarioTaxReduction // low-income reduction also exists
```

5. PAYROLL DATA — CPP, CPP2, EI (needed for credits, self-employed, and refund accuracy)

Parameter	2025	2026
CPP earnings ceiling (YMPE)	\$71,300	\$74,600
CPP basic exemption	\$3,500	\$3,500
CPP rate (employee)	5.95%	5.95%
CPP max (employee)	\$4,034.10	\$4,230.45
CPP2 second ceiling (YAMPE)	\$81,200	\$85,000
CPP2 rate / max (employee) ✓	4% / \$396.00	4% / \$416.00
CPP2 self-employed rate / max ✓	8% / \$792.00	8% / \$832.00
EI insurable max (MIE)	\$65,700	\$68,900
EI rate / max (employee)	1.64% / \$1,077.48	1.63% / \$1,123.07

```
cpp = 0.0595 × clamp(pensionableEarnings - 3500, 0, YMPE - 3500)
cpp2 = 0.04 × clamp(pensionableEarnings - YMPE, 0, YAMPE - YMPE)
ei = eiRate × min(insurableEarnings, MIE)
// self-employed pay BOTH halves of CPP (11.9% / 8%); half is a deduction, base portion
// credit/deduction split: base CPP (4.95% part) → credit; enhanced (1% part + CPP2) → d
```

6. INVESTMENT INCOME FORMULAS

6.1 Capital gains

```
taxableCapitalGain = 0.50 × (proceeds - adjustedCostBase - sellingCosts)
// inclusion rate is 50% – the proposed 2/3 increase was CANCELLED; do not build it in
// capital LOSSES offset gains only; net losses carry back 3 yrs / forward indefinitely
// superficial loss rule: loss denied if same security bought within ±30 days
```

6.2 Dividends (Canadian) — gross-up and credit

```
// ELIGIBLE dividends (most public companies, T5 box 24/25):
grossedUp    = dividend × 1.38
fedDTC       = grossedUp × 0.150198
// NON-ELIGIBLE (small business, box 10/11):
grossedUp    = dividend × 1.15
fedDTC       = grossedUp × 0.090301
// provinces have their own DTC rates – parameter table per province
// interest & foreign dividends: no gross-up, fully taxable at marginal rate
```

7. REGISTERED ACCOUNTS — LIMITS AND RULES

Account	2025	2026	Rules the engine must enforce
RRSP	18% of prior-year earned income, max \$32,490	max \$33,810	room accumulates (carry-forward from CRA Notice of Assessment); contributions in first 60 days of a year deductible against prior year; over-contribution buffer \$2,000 then 1%/month penalty; deduction can be deferred to a future higher-income year (optimizer opportunity)
TFSA	\$7,000	\$7,000	no deduction, growth tax-free; room accumulates from age 18/2009; withdrawals restore room next Jan 1 (re-contributing same year = over-contribution, 1%/mo)
FHSA	\$8,000/yr	\$8,000/yr	\$40,000 lifetime; deduction like RRSP and tax-free withdrawal for first home (best of both); unused annual room carries forward max \$8,000; 15-year max life; must be first-time buyer to open
RESP	\$50,000 lifetime/child		CESG grant = 20% of first \$2,500/yr = \$500 free money (optimizer nudge)

8. REFUND ESTIMATION (the number on your dashboard)

```
estimatedRefund = taxWithheld // Σ T4 box 22, T5 withholding
                  + instalmentsPaid
                  + refundableCredits // CWB, eligible educator, etc.
```

```
        - totalPayable                                // from §1
// CONFIDENCE: show a range, not a point. Drivers of uncertainty:
//   missing slips (AFR pull incomplete), unentered deductions, pending RRSP (first-60-d
// display: "Estimated refund $X - $Y" until all declared sources are entered
```

Why withholding $\neq$ tax owed (and refunds exist): employers withhold per-paycheque as if that pay rate ran all year, ignoring RRSP contributions, medical, donations, and multi-job effects. Two jobs = each withholds as if it's the only income = usually a balance **owing**. Detect the multiple-T4 case and warn the user early — it's the #1 "surprise bill" scenario.

9. MARGINAL vs EFFECTIVE RATE (dashboard metrics)

```
marginalRate = fedRate(taxableIncome) + provRate(taxableIncome)    // rate on the NEX
effectiveRate = (fedTax + provTax) / totalIncome                    // average on ALL
// example, ON 2025 @ $80,000 taxable: marginal = 20.5% + 9.15% = 29.65%
// note for accuracy: marginal rate near credit phase-out zones (BPA clawback, age amoun
// OAS clawback) is HIGHER than the bracket rate – compute marginal numerically:
trueMarginal = tax(income + 100) - tax(income) / 100              // robust method,
```

10. SIMULATOR FORMULAS (your differentiator)

10.1 RRSP contribution simulator

```
// NEVER use flat "contribution × marginalRate" – wrong when crossing brackets. Do this:
taxSaving = totalTax(taxableIncome) - totalTax(taxableIncome - contribution)
// where totalTax = fed + prov, full pipeline both times. Also recompute income-tested
// benefits (CCB, GST credit, OAS clawback) – RRSP deductions can INCREASE benefits;
// showing that combined number is a feature competitors don't have
alsoShow: refundPerDollar curve, room remaining, over-contribution warning
```

10.2 FHSA simulator

```
same taxSaving formula as RRSP (it's a deduction), capped at min(8000 + carryforward, 40
comparison view: FHSA vs RRSP-HBP vs TFSA for a down-payment goal
```

10.3 Capital gains timing simulator

```
taxThisYear = totalTax(income + 0.5 × gain) - totalTax(income)
taxNextYear = totalTax(projectedNextIncome + 0.5 × gain) - totalTax(projectedNextIncome)
show: difference, plus December/January split option (half the gain in each year
      uses two sets of brackets – legitimate and material for large gains)
```

10.4 Donation pooling simulator

```
// carrying forward 5 yrs & combining spouses beats claiming $150/yr repeatedly,
// because only amounts OVER $200/yr earn the 29% rate:
annualClaim = 5 × (lowestRate × 150) // five small claims
pooledClaim = lowestRate × 200 + 0.29 × (750 - 200) // one pooled claim
```

11. TAX HEALTH SCORE — PROPOSED FORMULA (your own metric — you define it)

```
score = ∑ weightedComponents, scaled to 0-100:
  25 pts registeredAccountUsage = 25 × usedRoom / availableRoom (RRSP+FHSA+TFSA ble
  20 pts deductionCapture        = 20 × claimedDeductions / detectedEligible
    (medical, donations, union dues, childcare, WFH-detailed, tuition)
  15 pts withholdingAccuracy     = 15 × (1 - |refundOrOwing| / totalPayable), floor 0
```

(big refund = interest-free loan to CRA; big owing = planning failure — both s

15 pts filingHygiene = on-time history, no missing slips vs AFR, instalmen

15 pts incomeStructure = credit for tax-efficient asset location

(interest inside registered accounts, dividends/gains outside, income splittin

10 pts planningEngagement = simulators run, RRSP deadline met, carry-forwards t

// Every point lost MUST map to a concrete action card ("Contribute \$X to RRSP: +N pts,

// — the score is only defensible if its dollar figures come from THIS engine, not marke

12. INCOME-TESTED CLAWBACKS (needed for accurate marginal rates)

Program	2025 parameters	Formula
OAS recovery tax ✓	2025: \$93,454 (recovery period Jul 2026–Jun 2027) 2026: \$95,323 (Jul 2027–Jun 2028)	$\text{clawback} = 0.15 \times \max(0, \text{netIncome} - \text{threshold})$, to 100% of OAS. Rate confirmed at 15%.
Canada Child Benefit	base \$7,997/child <6, \$6,748 ages 6–17 (2025–26 year)	reduced by family-income tiers — implement CRA's published phase-out table; RRSP deductions raise CCB (simulator hook)
GST/HST credit & CWB	CRA tables	refundable; phase in/out by family net income — table-driven, revisit annually

13. THINGS YOUR DEMO IMPLIES THAT ARE WRONG OR DEAD — FIX BEFORE CODING

- **Work-from-home \$2/day flat method is DEAD** (ended after 2022). WFH now requires the **detailed method**: employer-signed T2200 + T777 form, claiming $\text{workspaceArea}/\text{homeArea} \times \text{eligibleHomeCosts}$ (rent, utilities; commission employees add more; **mortgage principal never eligible**). The demo's "+\$180 WFH" card must be built on this, not the flat rate.
- **Consumer carbon rebate (CCR) ended April 2025** — no rebate card in the product.
- **Capital gains inclusion stays 50%** — the announced 2/3 increase was cancelled; don't build remnants of it.
- **The lowest federal rate is 14.5%/14%, not 15%** — any calculator copied from a pre-2025 tutorial will be wrong on every bracket-1 dollar and every credit.

14. HOW TO STRUCTURE THIS IN CODE

- **Everything above is config, not code.** One `tax_engine_params.json` keyed by `year` → `jurisdiction` → `parameter` (companion file provided). The formulas never change annually; only the tables do. Annual update = new JSON block + rerun test suite.
- **Money math:** compute in cents (integers) or a decimal type — never floating point. CRA rounding: per-line rounding to the cent; final balances to the dollar follow form rules.

- **Test harness from day one:** build ~50 personas (single T4; two T4s; T4+RRSP; self-employed; investor with gains/dividends; senior with OAS; parent with CCB; each province) and lock expected outputs. When CRA's certification test returns arrive, they slot into the same harness. Cross-check personas against an existing certified product's results while developing.
- **Every computed number carries provenance:** store which formula + which parameters produced it, so the UI can show "why" (trust feature) and support can debug (liability feature).

15. VERIFICATION LEDGER

15.1 Confirmed against CRA published pages (13 August 2026)

Item	Source page
Federal brackets, 2025 and 2026	tax-rates-brackets/current-year + last-year
All 12 provincial/territorial bracket sets, both years	same two pages
BPA max and min, both years; phase-out range	indexation adjustment page
Canada employment amount, age amount, medical cap, disability amount, spouse amount — both years	indexation adjustment page
Indexation factors (2.7% for 2025, 2.0% for 2026)	indexation adjustment page
CPP: YMPE, exemption, rate, maximum — both years	CPP contribution rates page
EI federal and Quebec: MIE, rate, maximum — both years	EI premium rates page
RRSP dollar limits (incl. 2027 = \$35,390), TFSA, YMPE/YAMPE	MP/RRSP/TFSA limits page
OAS recovery threshold and 15% rate, both years	OAS recovery tax page
CPP2: YAMPE, 4% employee rate, employee/employer max, 8% self-employed rate and max — both years	CPP2 rates and maximums page (calculating-deductions/making-deductions)
Dividend gross-ups (38% / 15%) and federal DTC factors (15.0198% / 9.0301%)	cross-checked, 2025
Capital gains inclusion rate 50%	confirmed — 2/3 proposal cancelled

15.2 Still unverified — do not ship a user-facing number using these

- **Provincial basic personal amounts** (all provinces, both years) — provincial tax is wrong without them
- **Provincial dividend tax credit rates** (all provinces)
- **Ontario surtax thresholds** (2025 and 2026) and the Ontario low-income reduction / LIFT credit
- **Age amount income threshold** — the point at which the 15% reduction begins
- **Donation tier rates** as they apply after the July 2025 rate cut
- **FHSA and RESP** parameters
- **Home buyers' amount**
- **CCB, GST/HST credit, Canada Workers Benefit** phase-out tables — not modelled in this document at all
- **British Columbia's 2026 lowest rate** (5.6% vs 5.06%) — flagged as a probable real rate change, needs confirmation

On the word "legal." Every figure in §15.1 now matches what the CRA publishes. That makes the data *accurate*; it does not make the software *compliant*. Legal authority to file returns comes from NETFILE certification — passing the CRA's own test returns each year — and nothing in this document substitutes for that. Treat this as the engineering spec you build against, then let the CRA's test suite be the judge.

Prepared August 2026; verification pass completed 13 August 2026 against CRA-published pages. Every figure must be re-validated each November when CRA publishes the following year's indexed amounts, and again before each certification cycle. This document is an engineering reference, not tax advice.